

SMART INDIA HACKATHON 2025



TITLE PAGE

- Problem Statement ID – 25010
- Problem Statement Title - Smart Crop Advisory System for Small and Marginal Farmers.
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Software
- Team ID- 71101
- Team Name – Hack Harvesters



1. Detail Explanation

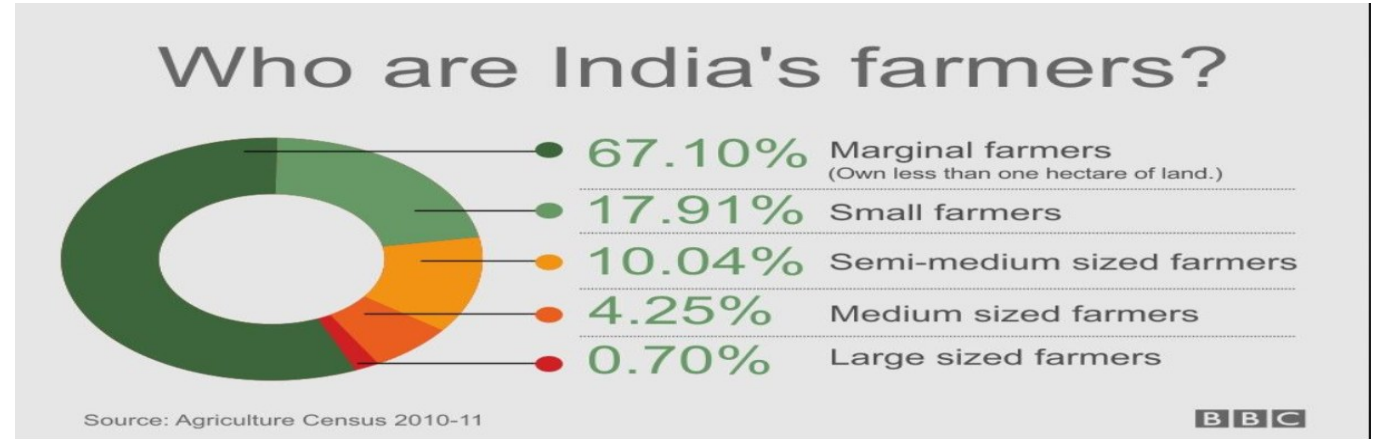
- Mobile-based Crop Advisory Website for small and marginal farmers.
- Real-time weather updates.
- Chatbot support.
- Information on government schemes/subsidies.
- User-friendly interface designed for easy use, even for farmers with limited literacy.

2. How It Addresses the Problem :

- Offering **instant expert advice**
- Providing **region-specific crop recommendations** based on soil, weather, and season.
- Ensuring **confidential, accessible, and affordable guidance** without dependency on middlemen.
- Empowering farmers to make **informed decisions** that increase yield and reduce losses.

3. Innovation and Uniqueness of the Solution

- AI + Expert Hybrid Advisory:** Combines artificial intelligence with human.
- Farmer-Centric Design:** Simple, multilingual, and voice-enabled.
- End-to-End Support:** Covers the full farming cycle — from seed selection to harvesting tips.
- Integration with Government Schemes:** Direct access to information on subsidies



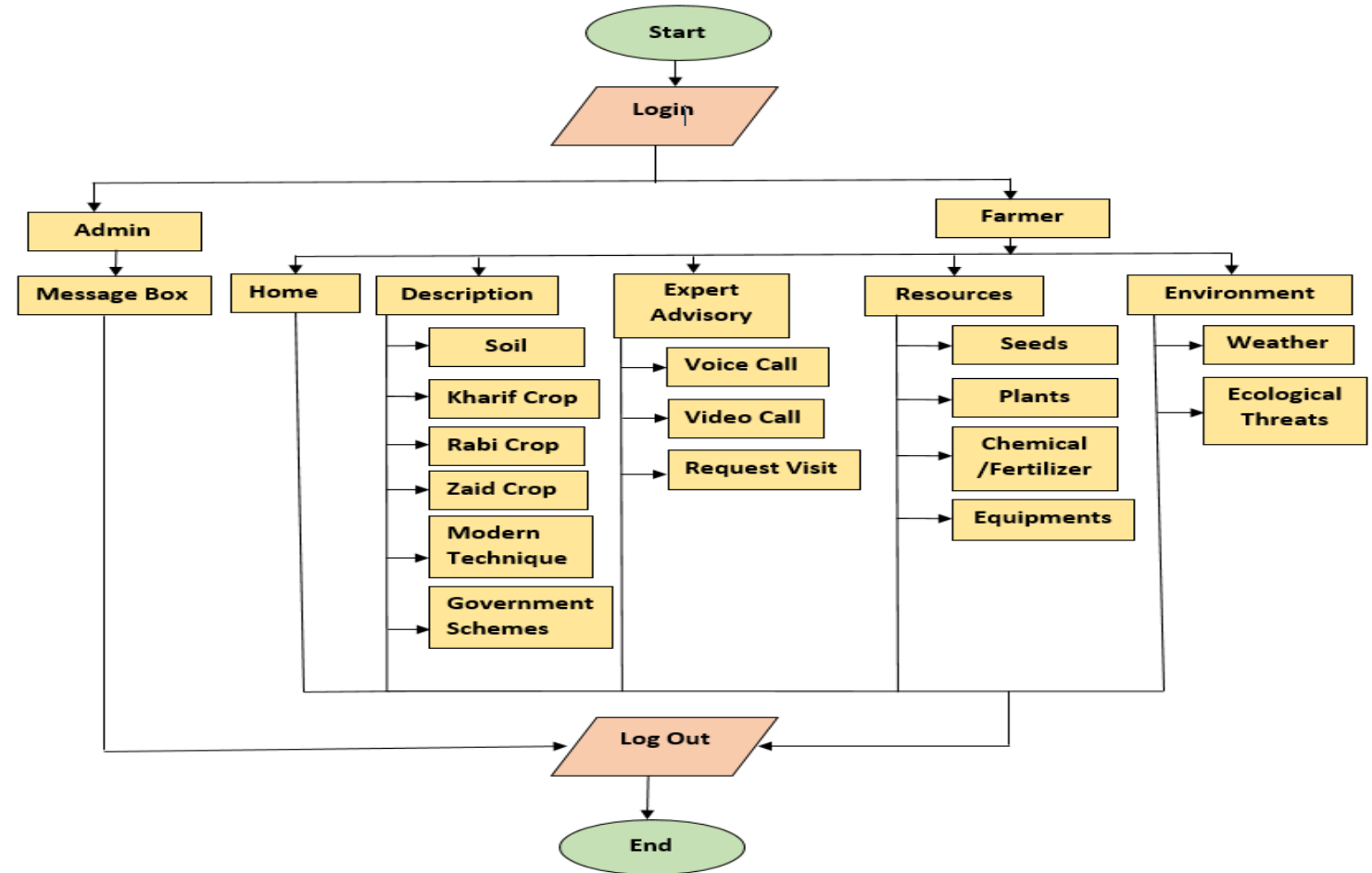
PROTOTYPE



1. **Frontend Languages:** HTML5, CSS3, JavaScript
2. **Frameworks/Libraries:** React.js, Bootstrap

2. **Backend (Server) Languages:** Python (Django), Node.js (Express)
Database: MySQL, MongoDB
Authentication: JWT / OAuth

3. **AI/ML Libraries:** TensorFlow, PyTorch, Scikit-learn, OpenCV
4. **Cloud & Hosting Platform:** Google Cloud
5. **APIs & Integrations:** Weather (OpenWeather, IMD), Government schemes, Agri-market prices
6. **Hardware (Future Scope):** Smartphone/computer (basic access), IoT devices: soil sensors, pH meters, weather stations



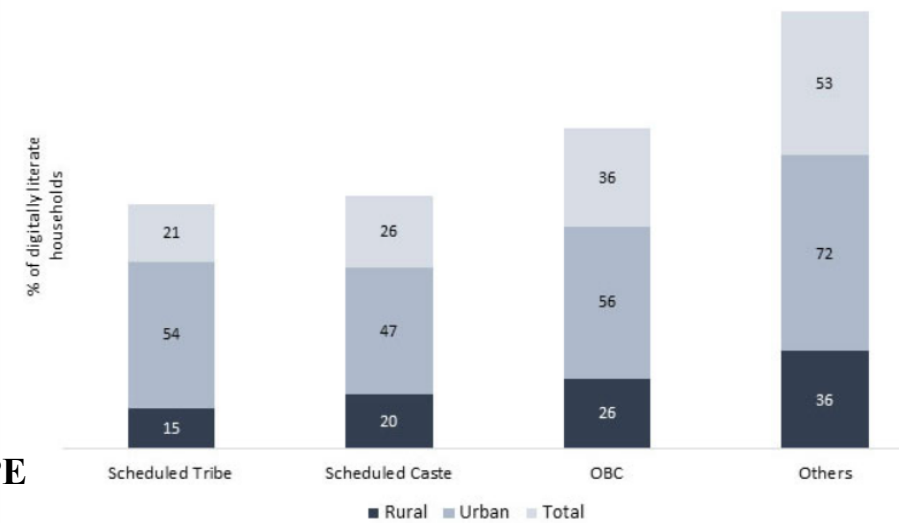
Prototype link - <https://youtu.be/vqORNXzhB3s?si=uQBgPC0wqlod-cHB>

1. Analysis of the Feasibility of the Idea

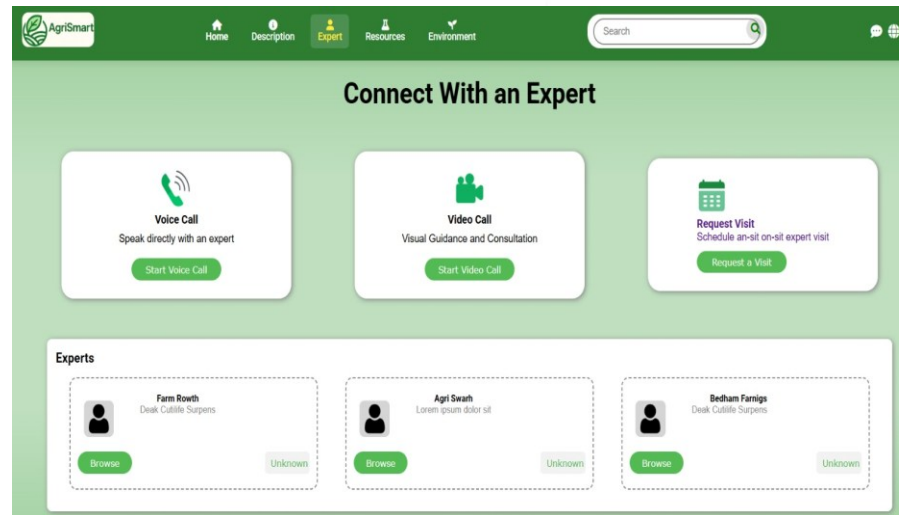
- Technical Feasibility
- Operational Feasibility
- Scalability Feasibility
- Sustainability Feasibility
- Adoption Feasibility
- Economic-Feasibility

3. Strategies for Overcoming These Challenges

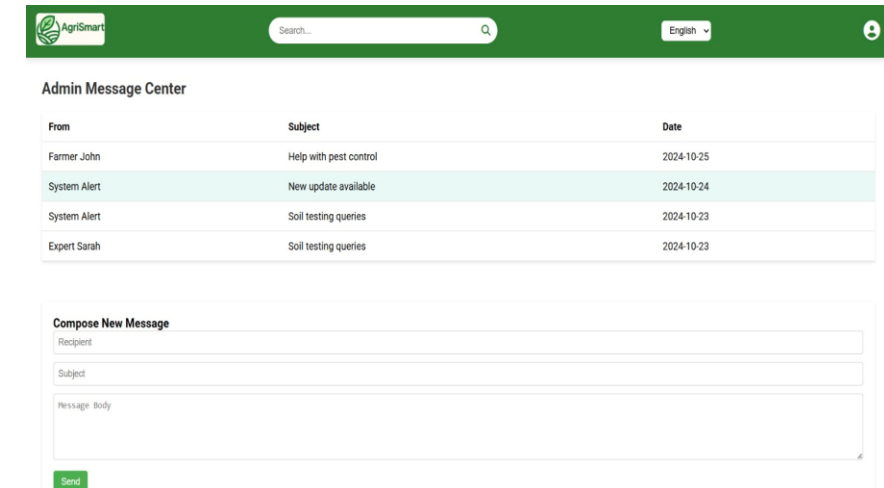
- **Digital Literacy:** Workshops & tutorials in local languages.
- **Offline Access:** Limited features, sync when online.
- **Trust Building:** Partner with agri officers, NGOs & cooperatives.
- **Sustainability:** Support via govt. schemes & agri-tech collaborations



PROTOTYPE



PROTOTYPE



2. Potential Challenges and Risks

- Digital Literacy
- Internet Connectivity
- Data Accuracy
- Trust Issues
- Financial Viability
- Language Barriers
- Device Affordability
- Privacy Concern

1. Potential Impact on the Target Audience :

- Empowers small and marginal farmers with timely, reliable, and personalized crop advisory.
- Increases awareness about modern farming techniques and government schemes.
- Improves decision-making in crop selection, pest control, and resource utilization.
- Enhances farmer confidence by reducing dependency on middlemen and unverified information sources.

2. Benefits of the Solution :

Social Benefits:

Reduces stress among farmers by offering instant support for farming challenges.

Promotes digital inclusion by providing easy-to-use, multilingual, and voice-enabled features.

Builds stronger farmer communities through shared knowledge and expert-farmer interaction.

Economic Benefits:

Increases crop yield and productivity through better farming practices.

Reduces losses caused by pest attacks, diseases, and unpredictable weather.

Enhances farmer income by suggesting profitable crops and connecting with markets.

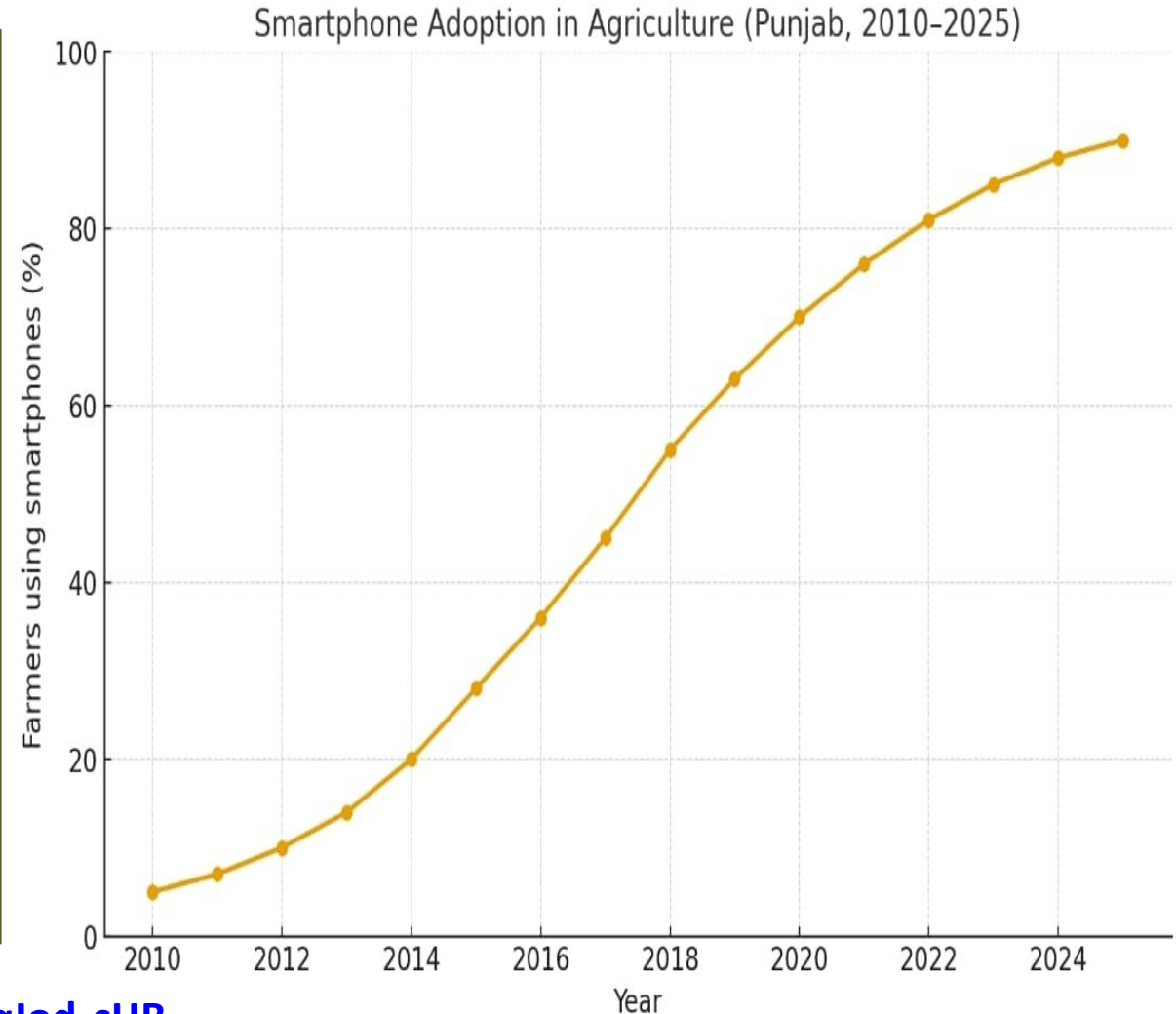
Lowers costs by optimizing fertilizer, pesticide, and water usage.

Environmental Benefits:

- Encourages sustainable farming practices, reducing overuse of chemicals.
- Promotes crop diversification to maintain soil fertility.
- Helps in efficient water management through weather-based irrigation advice.
- Reduces environmental impact by minimizing resource wastage.



- **Saagu Baagu (Telangana's AI4AI Initiative):** Farmers saw income doubling, yield increases, reduced pesticide & fertilizer usage through advisory, soil testing, quality AI tools. [World Economic Forum](#)
- **APAIMS 2.0 (Andhra Pradesh):** Digital agriculture platform providing plot-level pest/disease alerts, personalized advisories, real-time notifications. [The Times of India](#)
- **Plantix:** An existing crop advisory app using image-based disease detection and recommendations. [Wikipedia](#)
- Problems with **data availability & quality**, poor infrastructure in rural areas, low smartphone penetration, cost barriers, and lack of trust in AI-generated advice. [India Today](#)
- **Low Digital Literacy & Trust Issues:** Many smallholders struggle with using digital tools; trust & privacy concerns limit adoption [Springer Link](#)
- **Infrastructure & Training Barriers :** Lack of internet, devices, and farmer training reduce the effectiveness of digital extension services [MDPI Sustainability](#).



Prototype link - <https://youtu.be/vqORNXzhB3s?si=uQBgPC0wqlod-cHB>